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Predicting solar photovoltaic generation impacted by severe wildfire smoke

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Predicting solar photovoltaic generation impacted by severe wildfire smoke

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Fenya Bartram^{1,3}, Bo Yuan^{2,3} and K Max Zhang^{2,*} ¹ Department of Biological and Environmental Engineering, Cornell University, Ithaca, NY, United States of America² Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, NY, United States of America³ These authors contribute equally.

* Author to whom any correspondence should be addressed.

E-mail: kz33@cornell.edu**Keywords:** PV systems, solar energy generation, wildfire smoke, machine learning, renewable energySupplementary material for this article is available [online](#)**Abstract**

The negative impact of wildfire smoke on solar photovoltaic (PV) generation by reducing the amount of solar irradiance reaching the modules has been observed worldwide. However, the predictive capability to capture the impact on solar electricity production still needs to be improved. For example, in the summer of 2023, smoke from Canadian wildfires spread to the northeastern U.S., impacting solar PV output in the region. The New York Independent System Operator (NYISO) day-ahead forecasts for this period significantly overpredicted PV output. This paper presents novel machine learning-based models for predicting the hourly solar capacity factor, focusing on improving predictive performance during periods of severe wildfire smoke. The results demonstrate a R^2 value of up to 0.85 for the severe wildfire periods (aerosol optical depth (AOD) above the 99.99th percentile) from our models, significantly outperforming NYISO's R^2 value of 0.50 across six load zones included in the analysis. The greatly enhanced performance arises from two innovations. First, we adopted a series of data products, newly available in the public domain, from the high-resolution rapid refresh smoke (HRRR-Smoke) weather forecasting system. These include predictions of the AOD and the downward shortwave radiation flux incorporating aerosol impacts. Our study marks the first time the HRRR-Smoke wildfire AOD product has been used in solar electricity forecasts. Second, we employed upsampling strategies to address the data imbalance issues due to the inherently infrequent nature of wildfire events. As the data products are publicly available, our methodology can be readily adopted by power system operators to enhance predictions of solar electricity production during periods of wildfire smoke, ensuring the reliability of power grids with high penetration of solar energy.

1. Introduction

The frequency of wildfires in the United States is expected to increase due to the hotter and drier conditions driven by climate change [1]. Wildfire smoke poses a significant challenge to the reliability of solar photovoltaic (PV) energy production. Studies have shown that the smoke from wildfires reduces solar energy output, leading to fluctuations that can compromise grid stability. For example, wiggling PV output causes frequency instability in power grids due to sudden changes in PV generation [2]. A California-focused study found that monthly reductions in PV

power generation reached 5% during the 2020 wildfire season and could rise to 6% in the future as wildfire activity intensifies with climate change [3]. These reductions could lead to a 64% increase in monthly operating reserve requirements, significantly raising the operational cost of the power grid. Similarly, the 2019–2020 Australian Black Summer Wildfires were led to a mean power reduction rate of approximately $13 \pm 2\%$ per $100 \mu\text{g m}^{-3}$ of $\text{PM}_{2.5}$ [4]. Given that both solar energy penetration and wildfire frequency are expected to rise in the coming years, it is crucial to advance our predictive capabilities of wildfires' impact on solar power production.

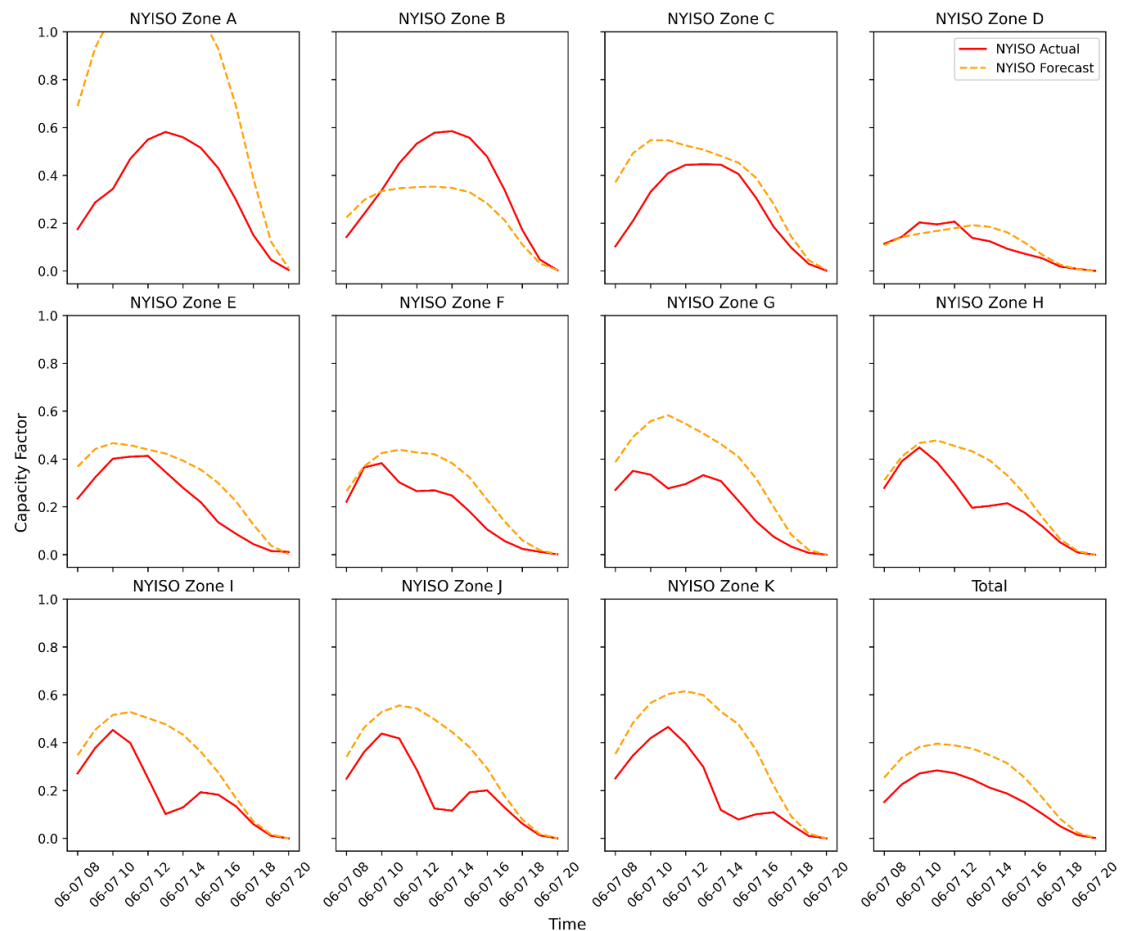


Figure 1. NYISO actual and day-ahead forecast solar CF for 7 June 2023, the day with the most severe wildfire impacts for load zones A–K (see figure 2). (Note: the CF above 1 for zone A is due to the forecast production exceeding the maximum capacity reported by NYISO.).

Several models have been developed to improve solar irradiance estimates or PV power production by including variables indicative of wildfire smoke as inputs. These include a geospatial PV capacity derating model for California, where wildfire smoke is represented by aerosol optical depth (AOD), a dimensionless number indicative of extinction of solar beams as a result of dust and haze, from historical weather reanalysis and satellite imagery [5]. Another regression model, developed for the Western U.S., relates daily PV production to observed $PM_{2.5}$ and historical weather data [6]. Similarly, in Washington State, a multivariate linear regression model trained on historical AOD and $PM_{2.5}$ air quality index (AQI) data was used to estimate solar irradiance [7]. Furthermore, absorption radiation index was employed to examine how different PV cell technologies respond to wildfire smoke using measured solar spectral data from a Montana wildfire in early September 2022 [8]. A random forest model (RFM) was also developed to determine the effects of wildfire smoke on solar PV production in the Western U.S., using station-based $PM_{2.5}$ and satellite weather data on a daily timescale [9].

Various techniques have been developed to estimate the effects of wildfire smoke on solar energy production. Nevertheless, all the studies reviewed above rely on historical weather data and air quality observations. Solar energy forecasting under wildfire smoke impact, although essential to power grid operation, still remains a significant challenge. For instance, the hour-ahead forecasts used by the California Independent System Operator over-predicted solar production by 10%–50% during a nine-day period in the 2020 wildfire season due to not considering the impacts of wildfire smoke [10].

While the impacts of wildfires have not been commonly observed on the East Coast in the past, the severe smoke from Canadian wildfires in the summer of 2023 significantly affected solar output in the northeastern U.S. [11]. The existing models adopted by the New York Independent System Operator (NYISO), which operates the state's bulk power system, were found to substantially overpredict solar production during this period, as illustrated in figure 1. These existing models typically perform much better for days without wildfire smoke, as shown in figure S4. This underscores the need for

improved forecasting models that account for wildfire smoke.

The main contribution of this study is to demonstrate that the wildfire smoke-related weather forecast data products, recently available to the public, can greatly enhance the predictions of solar PV capacity factor (CF) during episodes of wildfire smoke. Specifically, we utilise high-resolution rapid refresh smoke (HRRR-Smoke) forecasts developed by the National Oceanic and Atmospheric Administration (NOAA), which include AOD data as an indicator of wildfire smoke alongside various weather and site-specific variables.

Furthermore, analysing these impacts in the Eastern U.S. presents particular challenges due to the historical rarity of impacts from wildfire events, which limits the availability of data for smoke-impacted periods, creating an imbalanced dataset. To address this challenge, we draw on a technique that has been used for various applications involving imbalanced datasets: upsampling the minority data. This involves augmenting the amount of data from the minority case, which can enhance the model's performance for these rare events. Techniques like the synthetic minority over-sampling technique [12] have been applied successfully in other fields, including regression-based applications like solar flare prediction [13]. In our case, the minority data corresponds to periods of high AOD, indicating times impacted by wildfire smoke. Other methods for increasing the accuracy of imbalanced datasets, such as downsampling of the majority case (non-wildfire period data), assignment of higher weights to data from the wildfire period, various combinations of these, and an EasyEnsemble algorithm, were also tested, but led to poorer model performance. The results are shown in S8, including tables S7–S12.

While many studies rely on daily timescales, such as [6, 9], which offer more straightforward predictions, our model operates on an hourly basis. This offers important advantages in terms of grid planning and operation, especially considering that wildfire smoke impacts often vary throughout the day with smoke's movement in the atmosphere. We develop and compare two types of machine learning models, a RFM and a Hist Gradient Boosting model (HGBM), with the latter capable of predicting CF for solar sites during severe wildfire periods (when AOD is above the 99.99th percentile) with an R^2 value of 0.85. This model can improve forecasting of solar energy performance during wildfires, helping ensure the reliability of power systems supplied by solar energy during wildfire events. We also employ SHAP analysis to gain insight into how our model captures the effects of high AOD on CF.

2. Methodology

2.1. Data sources and cleaning

We utilised several datasets in this analysis: the New York State (NYS) Energy Research and Development Authority (NYSERDA) distributed energy resource (DER) integrated data system [14], the Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2), HRRR-Smoke, and the NYISO behind the meter (BTM) solar data. The datasets are shown in table 1.

The NYSERDA DER integrated data system provides performance data for DERs in NYS, including site attributes, such as nameplate capacity and geographic location, as well as historical generation time series data. Time series data for the CF, defined in equation (1), along with each solar site's latitude and nameplate capacity, were obtained from this data system,

$$CF = \frac{\text{Actual output}}{\text{Maximum possible output based on nameplate capacity}}. \quad (1)$$

MERRA-2 provides historical weather reanalysis data assimilated from numerical weather simulations and satellite observations, enhancing its credibility [15]. Therefore, it was used as part of the data-cleaning process to determine the occurrence of wildfire events. HRRR-Smoke, which provides predictions of several meteorological variables up to 48 h ahead [16], was used for developing predictive models. The 18 h-ahead weather prediction data was used, as this is the longest leading hour for which data is available for each hour. Predictions for leading hours greater than 18 are only available for four certain hours of a

day, not suitable for continuous solar electricity production forecasting.

There are several numerical weather models that provide wildfire-related data, as discussed in references [17–20]. However, HRRR-Smoke has a finer temporal resolution (hourly) and spatial resolution (3 km coverage for the continental United States compared to many of these other models, such as GEOS 'Forward Processing', the Copernicus Atmosphere Monitoring Service, the Realtime Air Quality Modelling System [17], and the NOAA Global Ensemble Forecast System, Aerosols [20].

Table 1. Summary of datasets used in this study.

Data source	Variables	Usage
NYSERDA DER Integrated Data System	CF time series, latitude, nameplate capacity	CF time series analysis; Features (latitude, nameplate capacity), and dependent variable (CF) for predictive model development
MERRA-2 (reanalysis)	Surface net downward shortwave flux (SWGNT)	Data cleaning
HRRR-Smoke (forecast)	Aerosol optical depth (AOD) for smoke at 550 nm, Downward shortwave radiation flux (DSWRF), air temperature at 2 m above ground level (AGL), wind speed at 10 m AGL, total cloud cover, relative humidity at 2 m AGL	AOD time series analysis; Features for predictive model development
NYISO BTM Solar	Actual and day-ahead forecast solar production data	Comparing model performance

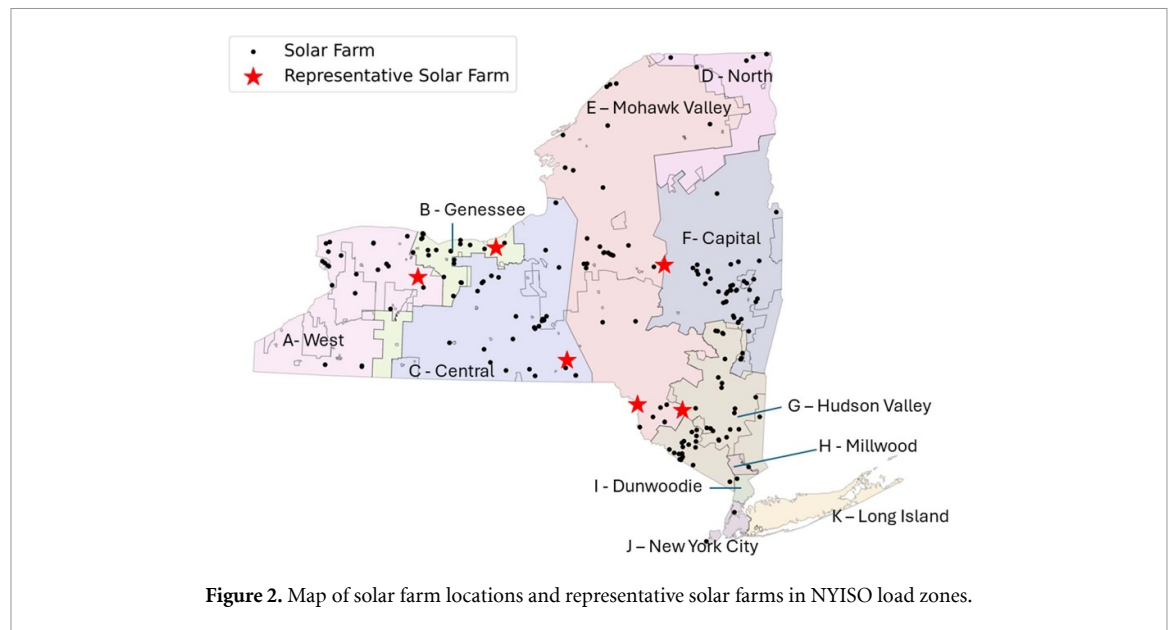
HRRR-Smoke is also publicly available, rather than being a custom-built NWP like WRF-solar, HRRR-Smoke is operational, rather than being an experimental product, such as air quality modelling and integration (FireWork experimental) [17]. Its availability and operational status allow for a much lower barrier to adoption than experimental datasets or custom-built NWPs, which require more time and expertise to implement. Leveraging the benefits of HRRR-Smoke, our work presents great opportunities to change existing practices to incorporate considerations of wildfire smoke impacts. Additionally, some other models are primarily intended for specific locations and thus less suitable for a study focused on NYS. For instance, the Air Information Report for Public Awareness and Community Tracking is focused mainly on the Pacific Northwest [19], and UCLA WRF-Chem only provides forecasts for the Western United States [17]. HRRR-Smoke also provides forecasts up to 18 h at hourly resolution, which are more useful for operational planning compared to models that only offer shorter-term forecasts. For instance, the Warn-on-Forecast System for Smoke generates 0–3 h ahead forecasts [18]. One of the other models, WRF-Chem, is used in the development of HRRR-Smoke.

This HRRR-Smoke data includes three variables related to wildfire smoke: the surface-level smoke mass density (MDENS), the total column-integrated smoke mass density (TCMDENS), and the AOD at 550 nm. These variables are influenced by wildfire smoke. The AOD variable was included in the analysis, while the TCMDENS was excluded because of its high correlation with AOD (the correlation coefficient was found to be 1). The MDENS variable was excluded due to unexplained high values for 2021, during the first year that the smoke capability was

implemented in HRRR [21]⁴. Among the weather variables provided by HRRR-Smoke is downward shortwave radiation flux (DSWRF), which quantifies the available solar resource and, therefore, plays an important role in predicting production. It is important to assess the forecast errors associated with the HRRR-Smoke forecasts. Figure S2(b) shows the Pearson correlation coefficients between 1 h and 18 h HRRR-Smoke AOD data for 7 June 2023, for each solar farm, indicating the model's predictive capability for this variable for times and locations impacted by wildfire smoke. Furthermore, table S1 shows a comparison between HRRR-Smoke and weather station data for several variables and between HRRR-Smoke and the MERRA-2 for AOD for the entire dataset and for only the severe wildfire period.

This study used hourly data from these datasets in the late spring and summer months (April, May, June, and July) when wildfires typically occur. Hours between 8 AM and 8 PM were selected to avoid including night time hours with zero solar output. The analysis focused on solar systems (above 1 MW) without energy storage. Sites were restricted to those for which data reporting began by 1 April 2021, and continued through 31 July 2023, to encompass the entire study period. 26 sites with more than 10 h of missing data between June 6 and June 8 were removed, as having data for this period was essential to understand the impacts of the Canadian wildfires. From the original 234 sites, these criteria led to the

⁴ The high values occurred in July of 2021, corresponding to a wildfire event, but were more than ten-fold greater than the values for the period in 2023 impacted by the wildfires in Canada. Other data sources as well as observation indicate that the 2023 wildfire smoke impacts were significantly more severe. Therefore, the MDENS variable was not used in model training.



inclusion of 208 solar sites, the locations of which are shown in figure 2.

It was necessary to clean the data to address a number of abnormalities that arose in the datasets. Specific hours for sites with illogically high CF and low MERRA-2 surface net downward shortwave flux (SWGNT) were removed, as these were presumed to be incorrect data. For example, hours with a CF greater than 0.6 while simultaneously having a SWGNT less than 5 W m^{-2} were removed from the analysis. Additionally, sites with zero production and SWGNT above the median were removed, accounting for zero production when the solar sites should have been operational, likely indicative of errors in data collection or events such as scheduled shutdowns. The strategy employed by Gilletly *et al* [9] was used to remove outliers by flagging and removing hours with a CF at least ten times greater or less than a consecutive hour. Each site hour represents data from one hour for a particular site. 83,763 site hours, or 8.5% of the data, were dropped from the analysis, reducing the number of site hours from 989,456 to 905,693.

2.2. AOD variable and wildfire period definitions

Elevated AOD was observed in NYS in June 2023 due to the wildfires in Canada. Figure S1 displays the distribution of the HRRR-Smoke AOD prediction data across the state on 7 June 2023, the day with the most severe wildfire impact. Figure S2(a) shows the average 1 h AOD for each solar farm for the same day. The AOD time series throughout the study period for the solar farm location with the highest maximum AOD on 7 June 2023, located in Zone C, is shown in figure 3. Data from other sites shows similar trends. The wildfire period was defined as hours when HRRR-Smoke 1 hour-ahead predicted AOD was above its 99th percentile for any solar site, indicated for this particular site by the yellow line in

figure 3. We use AOD one hour ahead instead of zero hour, because AOD is not calculated at the initialisation of the numerical weather simulation. The resulting dates were confirmed to align closely with several wildfires discussed in news sources, such as 16–23 July 2021, April 2023 and June 2023 [22–24], as well as periods when AQI in significant portions of New York is given as moderate, unhealthy for sensitive groups, unhealthy, or very unhealthy by [25]. The severe wildfire period was defined as hours when the HRRR-Smoke 1 h-ahead AOD was above its 99.99th percentile, indicated by the red line in figure 3. All these hours fall within the days of 5 June–8 June 2023, when smoke from Canadian wildfires impacted the Northeastern U.S.

Correlation matrices were used to depict the Pearson correlation coefficients between key variables during the severe wildfire periods, as well as for the entire dataset, as shown in figure 4. Unsurprisingly, DSWRF and CF exhibited strong positive correlations throughout the entire study period, with a correlation coefficient of 0.758, including the severe wildfire periods, with a correlation coefficient of 0.646. The DSWRF showed a negative correlation with AOD, which increased in magnitude from the baseline (−0.010) to the severe wildfire periods (−0.289). There was a slight negative correlation between AOD and CF. The negative correlation was minimal for the entire dataset (−0.010) because AOD was close to zero for most background hours without wildfire. However, the negative correlation became slightly larger (−0.043) during the severe wildfire periods. All of the p -values are less than 0.001, indicating the statistical significance of these relationships at the $\alpha = 0.001$ level. A Fisher's r -to- z transformation (Fisher's Z -test for correlation coefficients) was also performed in order to compare the differences in correlations between these two datasets, the one for the entire

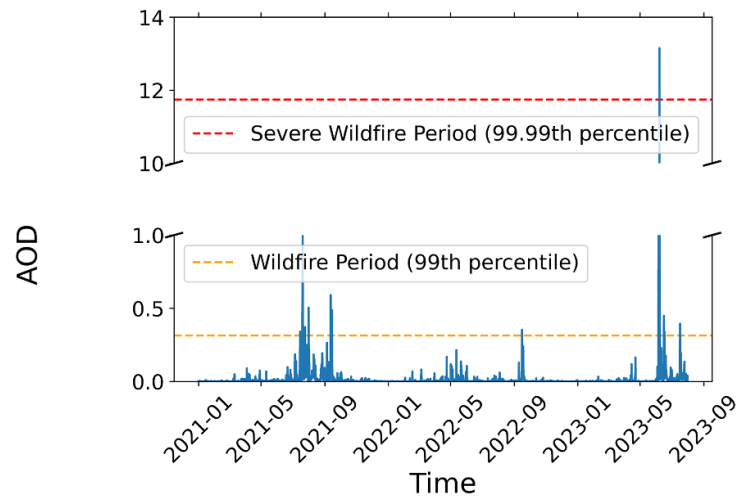


Figure 3. The 1-hour-ahead AOD time series throughout the study period, with wildfire and severe wildfire period thresholds, for a representative solar farm (with the highest AOD on 7 June 2023).

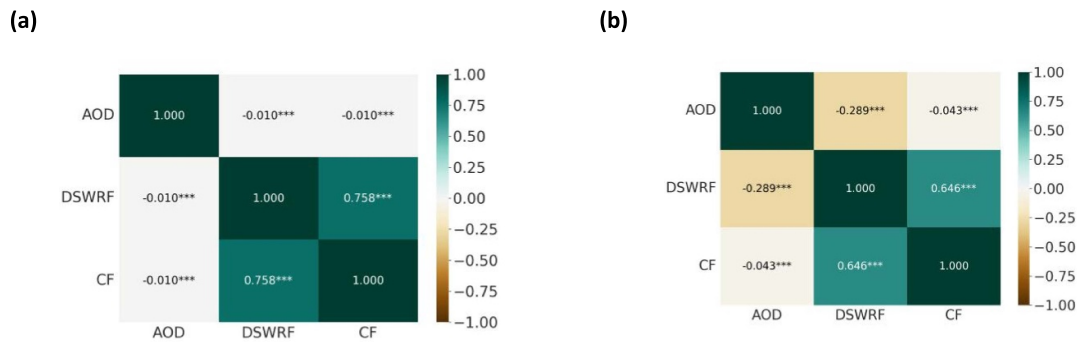


Figure 4. Correlation matrix for AOD, DSWRF, and CF for (a) the entire dataset and (b) the severe wildfire periods. *** indicates statistical significance at the 0.001 level.

period and the one for only the wildfire period [26]. The p -value was calculated to be 0.003 for the correlation between AOD and CF, and near 0 for the other two correlations, indicating that there are statistically significant differences between the two datasets for all three correlations. Thus, the AOD has a more substantial impact on the DSWRF and CF during the wildfire period than overall. This trend is generally consistent with the negative impact of AOD on both DSWRF and CF during wildfire periods observed in the literature. These findings, particularly the statistically significant correlation between AOD and CF, which is higher in magnitude during the severe wildfire period, justify the use of AOD as a variable in the predictive model to help represent the impact of wildfire smoke on CF, including the spatial and temporal variation in these impacts. Figure S3 shows the AOD and CF time series for representative solar farms in the six NYISO load zones with sufficient data to be included in the analysis, demonstrating the low CF generally observed corresponding to high AOD.

2.3. Model development

2.3.1. Features

The dependent variable for the machine learning models was the solar CF obtained from the NYISERDA DER integrated data system. The features used in the models are summarised in table 2. These include various HRRR-Smoke weather variables and site-specific variables from the NYISERDA DER integrated data system. The solar declination angle, the angle between the sun's direction and the equatorial plane of the Earth, was used to represent the seasonal variation of solar position. The solar azimuth angle, the angle between the ground projection of solar direction and the due south, was used to show the diurnal variation of solar position.

The solar declination angle (δ) was calculated using equation (2), where N is the day number on the Perpetual Julian Date Calendar. The solar azimuth angle (γ_s) was calculated using the *pvlb-python* [27] package, based on equation (3), where ϕ is the latitude, and ω is the hour angle (computed using equation (4)). Solar zenith angle (θ_z), the angle

Table 2. Summary of features used in machine learning models.

Feature	Units	Source
Aerosol optical depth (AOD) at 550 nm	Unitless	HRRR-Smoke
Air temperature at 2 m AGL (above ground level)	°F	HRRR-Smoke
Wind speed at 10 m AGL	m s ⁻¹	HRRR-Smoke
Total cloud cover	%	HRRR-Smoke
Downward shortwave radiation flux (DSWRF)	W m ⁻²	HRRR-Smoke
Relative humidity at 2 m AGL	kg kg ⁻¹	HRRR-Smoke
Solar site nameplate capacity	kW	NYSERDA
Latitude	Degrees North	NYSERDA
Solar declination angle (angle between the sun's direction and the equatorial plane of the Earth)	Degree	Equation (2)
Solar azimuth angle (angle between the ground projection of solar direction and the due south)	Degree	Equations (3)–(5) using <i>pvl-lib-python</i>

between a line from the sun and a vertical line, was calculated from equation (5),

$$\delta = -23.45 \sin \left[\frac{360 (284 + N)}{365} \right] \quad (2)$$

$$\gamma_s = \text{sign}(\omega) \cos^{-1} \left| \left(\frac{\cos(\theta_z) \sin(\phi) - \sin(\delta)}{\sin(\theta_z) \cos(\phi)} \right) \right| \quad (3)$$

$$\omega = \text{hour} \times 15 - 180 \quad (4)$$

$$\cos(\theta_z) = \sin(\delta) \sin(\phi) + \cos(\delta) \cos(\phi) \cos(\omega). \quad (5)$$

2.3.2. Hyperparameter tuning

The data was split 80–20 into training and testing sets of site hours. There were 724 554 site hours in the training set and 181 139 in the testing set. A sensitivity analysis was performed to consider the impact of the training/testing split on model performance, including the performance for the severe wildfire period. The results are shown in table S3. The data was stratified by month to ensure a similar distribution of months in the training and testing sets since the CF might vary between months. Two machine learning algorithms, a HGBM and a RFM, were developed and compared using the Scikit-learn package in Python [28]. A five-fold cross-validation with randomised search was implemented for hyperparameter tuning. A ten-fold cross-validation was also tested, as shown in table S6. However, the performance of the testing data was found to be slightly worse. For the RFM, the hyperparameters include the number of estimators, or trees in the forest, the number of features to consider when looking for the best split, the maximum depth of the tree, the minimum number of samples required to split an internal node, the minimum number of samples required to be at a leaf node, and whether bootstrap samples are used when building trees. For the HGBM, the hyperparameters included the number of iterations of the boosting process, the maximum number of leaves for each tree, the learning rate, and the minimum samples per leaf. The final hyperparameters selected for each model are summarised in tables S2 and S3.

2.3.3. Upsampling

These machine-learning models exhibited lower performance during the wildfire and severe wildfire periods, representing only about 5.1% and 0.8% of the total hours, respectively, than for all data, as shown in table S4. For instance, for the HGBM with AOD but without upsampling, the R^2 value is 0.882 overall, but declines to 0.874 and 0.820 for the wildfire and severe wildfire periods, respectively. The limited amount of data specific to these periods created an issue of an imbalanced dataset, a challenge inherent in examining the impacts of wildfire events in areas where such impacts are historically rare. This was addressed through an upsampling technique, which involved adding randomly resampled data from the wildfire period to the dataset. Different amounts of upsampling were tested, as shown in tables S6 and S7, and the most effective approach was to upsample data from the wildfire period by a factor of 10 for the RFM and 25 for the HGBM, meaning that the amount of data from the wildfire periods was increased by these respective amounts. We upsampled only the training data from the wildfire period after performing the train test split in order to prevent data leakage.

2.4. Model evaluation

The performance of each model was evaluated using three different metrics: the coefficient of determination (R^2), the root mean squared error (RMSE), and the mean absolute error (MAE). These performance metrics used align with those commonly selected for engineering applications. Naser and Alavi [29] argue for using multiple criteria in order to help overcome some of the limitations of traditional metrics. The three metrics can allow for a more robust analysis than a single metric by addressing each metric's limitations. For instance, while RMSE penalises larger errors more, making it more sensitive to outlier errors, MAE penalises errors equally and is thus less sensitive to outlier errors. MAE is also an unbiased estimator, in contrast to RMSE, which is biased, and MAE has a lower sample variance than

RMSE [30]. RSME is optimal for normal (Gaussian) errors, while MAE is best for Laplacian errors [30]. These metrics were calculated for the testing and training data for the entire dataset and the data from the wildfire periods (99th percentile of AOD) and severe wildfire periods (99.99th percentile of AOD). A baseline model without including the AOD variable was also developed to determine whether this variable improved model performance, particularly during the wildfire and severe wildfire periods.

The performance of the 18 h-ahead predictions for the models developed in this study was compared to that of the NYISO day-ahead predictions for the same period. The models trained on the entire dataset were applied to generate predictions using only the testing data from the zone, and these predictions were compared to the actual. The combined CFs for each NYISO load zone included in the analysis were calculated as the total generation divided by the total capacity. We only used the testing data from the severe wildfire periods for each NYISO load zone with at least five solar sites. A small number of days and hours were missing from either the predicted or the actual generation data from NYISO, so they were excluded when calculating the R^2 value to allow for the most accurate comparison. This analysis was repeated for three HGBMs, without AOD or upsampling, with AOD added as a feature but no upsampling, and with AOD included and upsampling used. We examined how each model's performance compared to the NYISO's day-ahead prediction.

SHAP (SHapley Additive exPlanations) analysis is a method based on game theory that is used to explain the output of machine learning models. This method was used to look at the contributions of different variables to the results. It was performed using the *Shap* package in Python [31] for both the data from the severe wildfire periods and a random sample (without replacement) of the same length from the entire dataset including non-wildfire days.

3. Results and discussion

3.1. Model performance

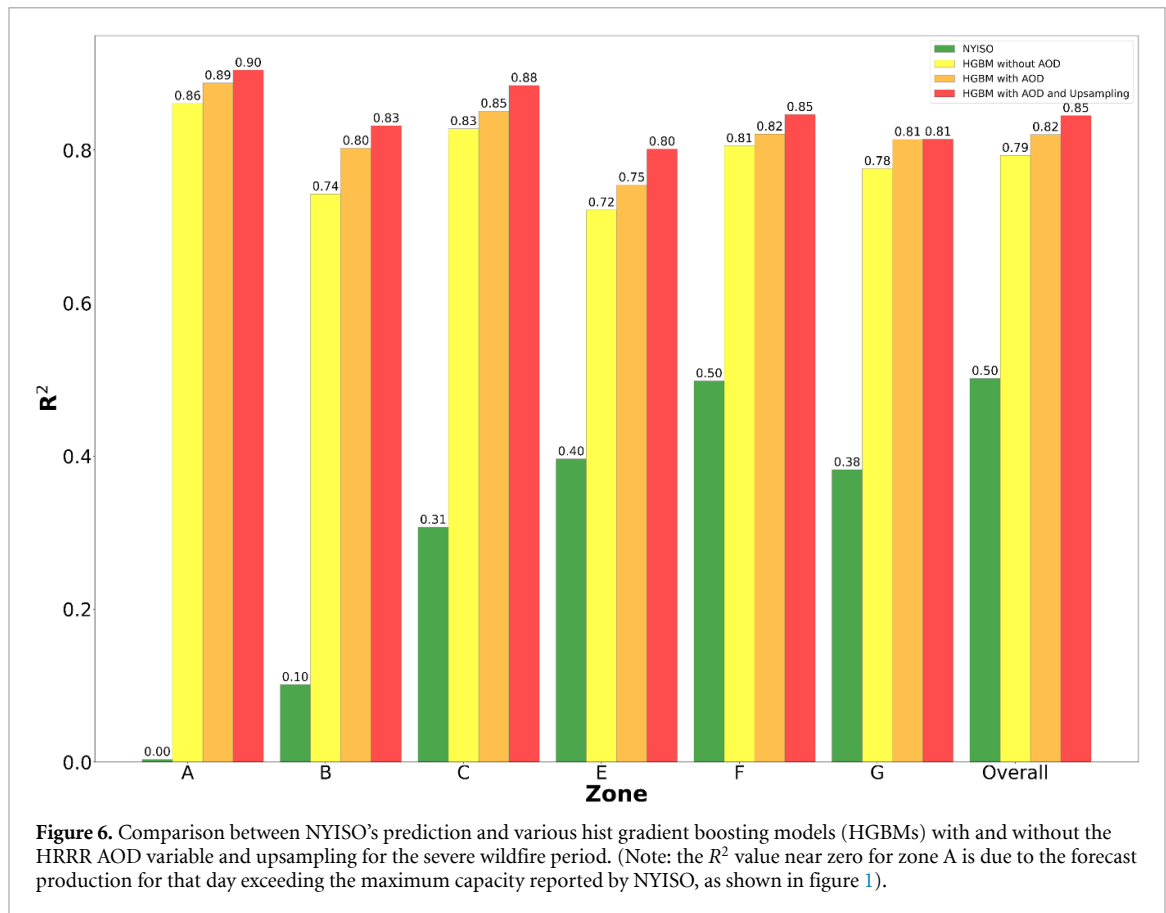
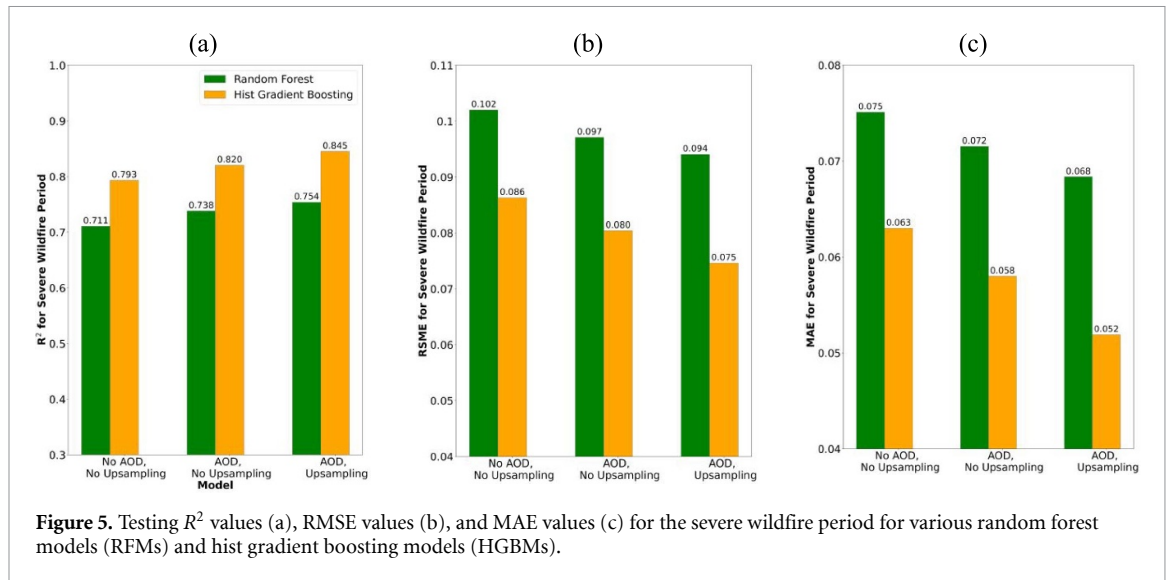
The R^2 , RMSE, and MAE values of NYISO's model and various RFMs and HGBMs trained in this study during different testing periods are summarised in table S4. Figure 5(a) displays the testing R^2 values for the severe wildfire periods for the two machine learning algorithms. Figure 5(b) shows the RMSE, and figure 5(c) shows the MAE, for the same models and data. The HGBM was found to be the most effective of the models tested, achieving the highest R^2 value and lowest RMSE and MAE values on the testing data, especially for the wildfire periods and severe wildfire periods. The inclusion of the AOD as a variable led to improved performance for both types

of machine learning models (RFM and HGBM), particularly during the wildfire and severe wildfire periods. Upsampling the data from the wildfire period led to further improvements for the wildfire periods and severe wildfire periods. The best-performing model during severe wildfire periods was an HGBM that included AOD and used upsampling, as shown by the right-most bar in figures 5(a)–(c). This model had a R^2 value of 0.88 for all the testing data (table S4). When tested on the data in wildfire periods, the R^2 value was around the same as tested on the entire dataset (0.89). However, when the HGBM was tested on data from the severe wildfire periods, the R^2 value decreased to 0.85. As shown in figure 5, HGBM outperforms RFM in all variations of model training schemes, namely without AOD, with AOD without upsampling, and with AOD and upsampling. Among the model training schemes shown, the last one, with AOD and upsampling, delivers the best performance for both RFM and HGBM.

3.2. Comparison to NYISO's prediction

Figure 6 shows the R^2 values of our models for severe wildfire periods on each load zone and the overall NYISO system, with and without AOD and upsampling. Figure S5 shows the same comparisons using RMSE rather than R^2 . Note that the HGBMs without AOD and upsampling already perform better than NYISO's incumbent model for the severe wildfire periods. This trend can be explained by using solar radiation predictions from the HRRR-Smoke DSWRF that account for the effect of aerosol in the numerical weather simulation. The inclusion of AOD led to improved performance across different zones. The models with AOD and upsampling showed the best performance. NYISO's solar CF predictions performed well for the entire dataset, with an overall R^2 value of 0.93, as shown in table S4, and R^2 values ranging from 0.87 to 0.90 for each load zone included in this analysis. In contrast, their performance decreased significantly during the severe wildfire period, with an average R^2 value of 0.50, ranging from 0.00 to 0.50 for the load zones, as shown in figure 6. Our model, on the other hand, was able to achieve a higher average R^2 of 0.85 for the severe wildfire period. Furthermore, the R^2 value during the severe wildfire period was higher than that of NYISO for each of the six load zones included in the analysis.

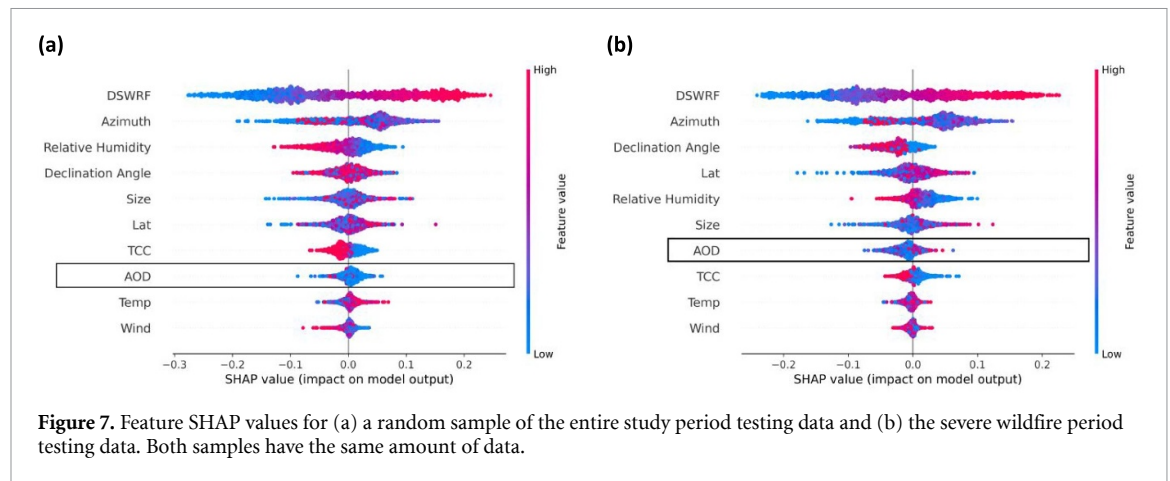
In addition to being accounted for explicitly through AOD, the wildfire smoke may impact other variables used in the model, particularly the DSWRF. This relationship is shown by the statistically significant negative correlation between AOD and DSWRF in figure 4, which increases as the AOD percentile increases. Thus, some of the improvements in our model may result from the fact that, unlike NYISO's predictions, which do not account for wildfire impacts, our model is based on weather data, which accounts for these. It can explain why the



model developed in this study was able to outperform NYISO during the severe wildfire period even when AOD was not included as a feature, with its explicit inclusion leading to further improvements. Even when the smoke-related variables, like AOD, are not explicitly used in the machine learning model, having DSWRF implicitly impacted by smoke also helps the prediction.

In order to quantify the benefits enabled by the proposed model, we estimated the potential savings from more accurate predictions by avoiding the

need to purchase additional reserves on 7 June 2023. NYISO's actual and predicted datasets were used to estimate the total amount of overprediction for each hour across zones. Using only the testing data for each site, the overprediction for the best-performing proposed model (HGBM with AOD and upsampling) was calculated and scaled by the ratio of the total capacity to the capacity of the included solar farms, to estimate the overprediction for the total capacity. Overprediction was assumed to incur a cost per MWh based on hourly spinning reserve prices



obtained from Energy Market & Operational Data NYISO [32]. Underprediction would be less costly, since the operator could curtail excess solar generation without needing to procure additional spinning reserves, so its cost was neglected for our analysis. Based on this approach, the proposed model would reduce the costs of spinning reserves by about \$38 000 for 7 June 2023. It is important to note that these benefits will be greater in the future, given the planned expansion of solar capacity. According to NYISO's System Resource Outlook [33], installed solar capacity is projected to increase by 41 475 MW by 2042, a more than 6-fold increase from its 2023 capacity of 7195 MW. Assuming benefits scale with the installed capacity, potential savings could rise to \$257 000 per day by 2042 under conditions similar to those on 7 June 2023. Thus, although constructing machine learning models involves computational costs, the resulting improvements in forecast accuracy could yield substantial operational savings, especially as solar capacity continues to expand.

3.3. SHAP analysis

The SHAP analysis further demonstrates that the AOD resulting from wildfire smoke has a greater negative impact on the predicted CF during the wildfire periods compared to the entire study period, which mainly consists of background hours without wildfire [34]. Figure 7 shows SHAP values for each feature, first for a random sample of the entire study period testing data (without replacement), and then for the severe wildfire period testing data. The colour bar represents the normalised value of the features. The x -axis shows a feature's SHAP values, measuring its impact on model output, evaluated at the data samples. When the model is tested on the sample of the whole dataset, solar irradiance DSWRF is shown to be the most important feature, followed by azimuth angle and relative humidity. The AOD variable is the eighth most important variable, being more important than only wind speed and temperature. Additionally, the SHAP values appear to average

around zero, indicating that the impact is generally minimal without an apparent direction.

However, for severe wildfire periods, the AOD rises to the seventh most important feature, which is now more important than total cloud cover. Also, most of the SHAP values are negative, indicating the negative impact of high AOD on CF. Note that the colour scale differs between the two figures since the AOD is higher for severe wildfire periods by definition (99.99th percentile of AOD). These observations imply that the model accounts for AOD's significant negative impact on the predicted CF during the severe wildfire periods, whereas it neglects AOD for times with no wildfire. This argument is consistent with the findings of improved performance of predictive models during the severe wildfire period when AOD is included as a variable. The SHAP analysis shows that the model developed in this paper is able to accurately capture the negative impact of AOD on CF during periods of elevated AOD resulting from wildfire smoke. In order to help understand these findings at a regional level, the SHAP analysis was also conducted for each NYISO load zone, as shown in figure S6.

4. Conclusion

In this paper, we demonstrated that incorporating wildfire smoke data significantly improves the performance of solar energy forecasting models during wildfire periods, making use of HRRR's new open-source smoke forecast product, HRRR-Smoke. Our machine learning model, trained using the 18 h-ahead AOD variable from HRRR-Smoke, performed particularly well during the severe wildfire periods. Upsampling data from wildfire periods alleviated the data imbalance problem caused by the rarity of wildfire smoke events on the East Coast, further enhancing the model's performance. As a result, our model predicted solar CF for each load zone with greater accuracy than NYISO's predictions during the severe wildfire periods. It also offered hourly forecasts—a key advantage over existing models based on larger timesteps, such as daily. The SHAP analysis provided

insights into the critical role AOD plays in CF prediction during severe wildfire periods, showing that the model effectively captures the impact of wildfire smoke on solar output.

Based on these findings, we recommend that grid operators account for the impacts of wildfire smoke, including both doing so explicitly through the inclusion of variables such as AOD and through the use of other variables, particularly solar irradiance, which are influenced by smoke in the forecast process. Techniques such as upsampling can further improve predictions during periods of high wildfire smoke, particularly in regions where these periods are uncommon. One potential strategy is to develop a specialised model for high-AOD periods, optimizing its performance with upsampling. By enabling more accurate prediction of solar output during wildfires, these recommendations can help support greater integration of solar energy, aiding decarbonisation efforts while reducing the power system's vulnerability to increased wildfires driven by changing climate.

While NYS is used as a case study, the method presented here may be applied to other areas, particularly in the Northeastern United States, where data and prior work on wildfire impacts are relatively limited due to the historical rarity of these impacts. The use of open-source weather data from HRRR-Smoke, which is available for areas throughout the continental United States, can allow this approach to be applied to other areas. The limited amount of data for solar performance during periods impacted by wildfire smoke presents a challenge for model development and testing. Limitations include the computational efforts associated with building machine learning models. However, there are also significant benefits which can be obtained from improved models, by avoiding the need to purchase spinning reserves, as described in the discussion section. Future work can focus on examining how these solar forecasting errors caused by wildfire smoke impact power systems. Additionally, more site-specific variables, such as module and inverter characteristics, may be integrated into future models in order to allow for greater accuracy given these variables' impacts on performance in general as well as on the extent to which this performance is degraded by wildfire smoke [8]. Furthermore, this study focused primarily on the impacts of wildfire smoke through impacting solar insolation reaching the panels, while the impact caused by soiling, or the build-up of particles on panels [9], was neglected in this analysis and should be explored in future research.

Data availability statement

HRRR-SMOKE data can be downloaded from <https://rapidrefresh.noaa.gov/hrrr/HRRRsmoke/>;

NYSERDA DER generation data can be downloaded from <https://der.nyserda.ny.gov/>; MERRA-2 reanalysis data can be downloaded from <https://gmao.gsfc.nasa.gov/reanalysis/> MERRA-2; NYISO BTM solar generation data can be downloaded from www.nyiso.com/reports-information.

All data that support the findings of this study are included within the article (and any supplementary files).

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ORCID iDs

Bo Yuan  <https://orcid.org/0000-0001-6359-4051>
K Max Zhang  <https://orcid.org/0000-0002-3324-6571>

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